第 1 章 广义函数与 Sobolev 空间

定义 1.1

记多重指标 $\alpha = (\alpha_1, \alpha_2, \cdots, \alpha_n)$, 其中 $\alpha_1, \alpha_2, \cdots, \alpha_n \geqslant 0$ 是整数,

$$|\alpha| = \sum_{i=1}^{n} \alpha_i, \quad \alpha! = \alpha_1! \alpha_2! \cdots \alpha_n!,$$

$$x^\alpha = x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_n^{\alpha_n} \quad (\forall x = (x_1, x_2, \cdots, x_n) \in \mathbb{R}^n),$$

$$\partial^\alpha = \partial_{x_1}^{\alpha_1} \partial_{x_2}^{\alpha_2} \cdots \partial_{x_n}^{\alpha_n},$$

而且如果 $\beta \leqslant \alpha$(系指 $\beta_j \leqslant \alpha_j, j = 1, 2, \cdots, n$), 则

$$\binom{\alpha}{\beta} = \frac{\alpha!}{\beta!(\alpha - \beta)!} = \binom{\alpha_1}{\beta_1} \binom{\alpha_2}{\beta_2} \cdots \binom{\alpha_n}{\beta_n}.$$

♣

1.1 广义函数的概念